Spectral–Spatial-Weighted Multiview Collaborative Sparse Unmixing for Hyperspectral Images

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***Abstract*— Spectral unmixing is an important task in hyperspectral image (HSI) analysis and processing. Sparse representation has become a promising semisupervised method for remotely sensed hyperspectral unmixing and incorporating the spectral or spatial information to improve the spectral unmixing results under a weighted sparse unmixing framework is a recent trend. While most methods focus on analyzing HSI by exploring the spatial information, it is known that hyperspectral data are characterized by its large contiguous set of wavelengths. This information can be naturally used to improve the representation of pixels in HSI. In order to take the advantage of the hyper spectral information as well as the spatial information for hyperspectral unmixing, in this article, we explore and introduce a multiview data processing approach through spectral partitioning to benefit from the abundant spectral information in HSI. Some important findings on the application of multiview data set in sparse unmixing are discussed. Meanwhile, we develop a new spectral–spatial-weighted multiview collaborative sparse unmixing (MCSU) model to tackle such a multiview data set. The MCSU uses a weighted sparse regularizer, which includes both multiview spectral and spatial weighting factors to further impose sparsity on the fractional abundances. The weights are adaptively updated associated with the abundances, and the proposed MCSU can be solved by the alternating direction method of multipliers efficiently. The experimental results on both the simulated and real hyperspectral data sets demonstrate the effectiveness of the proposed MCSU, which can significantly improve the abundance estimation results.**

***Index Terms*— Hyperspectral imaging, multiview learning, sparse regression, spectral partitioning, spectral–spatial-weighted unmixing.**

I. Introduction

DUE to the limited spatial resolution of sensors, together with microscopic material mixing and multiple scattering, mixed pixels are typically present in hyperspectral image (HSI) and severely degrade the processing and application of HSI [1], [2]. Therefore, hyperspectral unmixing (HU) has been an important technique for the exploitation of hyperspectral data, which aims at inferring the constituent spectra (endmembers) and their corresponding fractions (abundances) in each pixel of HSI [3]. Linear mixing model (LMM) has been widely used for HU as its computational tractability and flexibility, and it assumes that each observed mixed pixel in HSI is a linear combination of endmember signatures weighted by their corresponding abundances [4], [5]. Under this model, early unmixing methods based on geometry [6]–[8], statistics [9], [10], and nonnegative matrix factorization [11], [12] are all unsupervised algorithms and extract endmembers directly from HSI. Nevertheless, some of them [9]–[12] gain virtual endmembers without physical meaning, and others [6]–[8] require the presence of pure pixel in the image (this assumption is hard to guarantee in real HSI). In addition, estimating the number of endmembers in HSI required for these methods is also a difficult problem.

In order to tackle these issues, sparse unmixing [4], [13] has been proposed as a semisupervised approach, in which mixed pixels are expressed as the combinations of a small number of spectral signatures known in advance and available in a (potentially very large) spectral library, such as the well-known U.S. Geological Survey (USGS) library [14]. It amounts to finding the optimal subset of spectral signatures in the spectral library that can best model each mixed pixel in HSI, where it does not need to extract endmembers or estimate the number of endmembers in the scene. The sparse unmixing algorithm via variable splitting and augmented Lagrangian (SUnSAL) [13] has used classic l_1 regularizer on the abundance matrix, thus making use of the fact that a pixel is less likely to be mixed by a large number of spectral signatures. However, the high mutual coherence of spectral libraries limits the performance of SUnSAL. To mitigate it, the total variation (TV) regularizer is introduced to integrate the spatial-contextual information of neighboring pixels, and the SUnSAL-TV [15] algorithm is proposed. However, it may yield oversmoothness and blurred boundaries. Nonlocal sparse unmixing (NLSU) [16] adopts the nonlocal spatial information of abundance map for sparse unmixing and enables better unmixing performances with a more complex model, which limits its practical application. From different perspectives, the collaborative SUnSAL (CLSUnSAL) [17] introduces the $l_{2,1}$ regularizer and applies sparsity to the endmembers simultaneously for all pixels.

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A recent trend to address this issue is to incorporate the spectral or spatial information to improve the unmixing results under a weighted sparse unmixing framework. These algorithms penalize the nonzero coefficients of the abundance estimation matrix by introducing weighting factors [18]. In [19], an iterative weighted sparse regression problem with weighted l_1-norm is presented to enhance the sparsity of the abundance matrix. In the double-reweighted sparse unmixing (DRSU) [20] algorithm, dual weights are used to enhance the column sparsity of abundances and the sparsity along the abundance vector of each endmember. Meanwhile, spatially weighted factors are also used to interpret the spatial correlation information in sparse HU. Considering that endmembers are typically distributed in local spatial regions rather than uniformly distributed throughout the image, the local collaborative sparse unmixing (LCSU) [21] uses a spatial weight to maintain local collaborativity for sparse unmixing. In order to preserve the edge information in abundance estimation, [22] develops a spatial discontinuity weighted sparse unmixing approach. The discontinuity preserving (DP) strategy is designed to smooth the textures while retaining sharp edges in the abundance maps. To explore spectral and spatial information simultaneously, a spectral–spatial-weighted sparse unmixing (S^2WSU) method is proposed in [23] for HU. It uses a spectral weight to promote nonzero row vectors and a spatial weight to combine the spatial correlation information through a neighborhood system under the l_1 framework. Readers are also referred to the literature such as [24]–[28] for concurrent developments of weighted sparse unmixing in semisupervised HU. These weighted sparse unmixing methods show good potential for using spectral or spatial weights in two main aspects: the same or better unmixing performance and lower computational complexity. Similar to the spectral or spatial regularizers, these approaches explore spectral or spatial information in the estimated abundances; however, the weighted sparse unmixing methods only need one regularization parameter and exhibit much lower computational complexity.

In the abovementioned sparse unmixing methods, [13], [17], [19] and [20] use only spectral information to process the HSI pixels. Most of the sparse unmixing methods [15], [16], [21]–[23], [26]–[28] mentioned above are mainly developed to make use of the spatial-contextual information. As pointed out in [15] and [26], it is important to involve the spatial-contextual information in the processing of HSI. Meanwhile, it is known that HSI is characterized by its large contiguous set of wavelengths. It provides contiguous or noncontiguous 10-nm bands in the range 0.3–2.5 μm of the electromagnetic spectrum [4]. Utilizing the wealth of available spectral information, it is possible and needed to benefit from this hyperspectral information to further reduce the unmixing errors.

In order to take the advantage of the abundant spectral information, spectral analysis or spectral partitioning has been widely used in HSI classification. Different from traditional band selection methods, spectral partitioning, which provides an approach to cope with the high dimensionality of HSI, utilizes all spectral bands of the input data for the subsequent image processing by generating multiple views of the original HSI. Spectral partitions can be acquired by selecting a subset of spectral bands that effectually subsamples the original spectral signatures while reserving their dominant characteristics. As shown in [29], a spectral partitioning method is proposed by partitioning the original bands into several highly correlated subgroups to achieve better performance on the principal component analysis (PCA) transformation. Reference [30] introduces a spectral and spatial partitioning approach and incorporates it into PCA-based compressive projection for HSI reconstruction. By using adaptive affinity propagation (AAP) [31], the spectral partitioning method in [32] first intelligently generates correlated spectral band clusters from the original spectral signatures. The bands are then reassigned from the clusters into new partitions to form a subsampling of the original bands. Similarly, utilizing an interband correlation matrix, the work in [33] divides the original spectral bands into nonoverlapping subsets of correlated bands, and several efficient fusion techniques are proposed. Differently, considering the physical characteristics of the imaging spectrometer, reference [34] proposes a new spectrometer-driven (SD) spectral partitioning strategy and obtains a significant gain in HSI classification performance.

After applying spectral partitioning, or multiview hyperspectral data, the performance of HSI classification has been significantly improved. While in HU, there are also some attempts to apply spectral partitioning techniques to efficiently utilize the rich spectral information in HSI. Ammanouil *et al.* [33] perform the unmixing on each partition and use this complementary information to enhance the original results. Meanwhile, several fusion techniques for unmixing are proposed, and the final estimated abundances have been improved. In [35], a spectral–spatial multiview data generation approach is proposed by dividing the original data set into several homogeneous regions with different spectral partitions. A multiview dictionary learning method is presented. Utilizing the spectral–spatial multiview data set, the proposed algorithm substantially improves the unmixing performance.

Based on the advantages observed by the use of multiview data set in HSI classification and unmixing, in this article, a multiview data processing approach is explored and introduced for sparse unmixing to benefit from the abundant spectral information in HSI. Meanwhile, we develop a new spectral–spatial-weighted multiview collaborative sparse unmixing (SSMCSU) algorithm, which constrains the abundances simultaneously from both spectral and spatial domains, to tackle such a multiview data model. The main contributions of this article can be summarized as follows.

1) We explore the application of multiview data processing methods in sparse unmixing through spectral partitioning, which is widely used in HSI classification, to benefit from the rich spectral information in HSI. In the field of sparse unmixing, we put forward the idea of multiview sparse unmixing for the first time. And, we make an important discovery that the estimated abundance matrices of virtual endmembers have large differences on different views of the data set. Two spectral partitioning approaches, i.e., AAP and SD spectral partitioning,

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are discussed, which relies on statistical principles and physical principles, respectively.

2) Based on the discovery, we develop a new SSMCSU model that uses a multiview spectral- and spatial-weighted sparse regularizer to further enhance the sparsity of the fractional abundances. Specifically, the multiview spectral weight enhances the sparsity of nonzero rows associated with true endmembers in a multiview collaborativite manner by calculating the similarity of endmembers in different views (which represents a completely original contribution of this article), whereas the spatial weight, with previous developments, combines the spatial-contextual information to promote sparsity through a neighborhood system.
3) With respect to computational complexity, we do not adopt the form of regularization terms, and based on the previous analysis, we believe that the weighted regularizer form is superior in terms of performance and complexity. The proposed MCSU can be efficiently solved via the alternating direction method of multipliers (ADMM) [36], with only one regularization parameter and much lower computational complexity.

The rest of this article is structured as follows. Section II introduces the background of sparse unmixing theory. In Section III, we present the multiview data construction approach and the proposed SSMCSU model. The experimental results and a detailed comparison with other state-of-the-art methods are shown in Section IV. Our experimental results on both simulated and real hyperspectral data indicate that the proposed MCSU exhibits very good performance, especially in noisy scenarios. Finally, we conclude in Section V with some remarks.

II. Sparse Unmixing

According to the linear sparse unmixing model, each mixed pixel in the HSI is a linear combination of only a few spectra in a spectral library known in advance. Let $A \in R^{L\times m}$ denote the spectral library including L spectral bands and m spectral vectors. Then, the observed HSI can be modeled as [13]

$$y = Ax + v \tag{1}$$

where y is the spectrum vector of a pixel in the HSI with L-bands, x is an abundance vector of the elements in library A, and v is an observation noise. The model is supposed to meet two physical constraints: $x_p \geq 0$ (x_p is the pth element of x) and $\sum_{p=1}^{m} x_p = 1$, which correspond to the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC), respectively [37]. Because of some criticisms on ASC [13], it is rarely adopted. For convenience, we rewrite (1) in a matrix form

$$Y = AX + V \tag{2}$$

where $Y \in R^{L\times n}$ is the HSI matrix with L spectral bands and n pixels, $X \in R^{m\times n}$ is the abundance matrix, whose elements correspond to abundance of m spectra in n mixed pixels, and $V \in R^{L\times n}$ is an observation noise matrix.

The number of endmembers contained in the original HSI is generally quite small compared with m (usually very large), i.e., X is sparse. Usually, we represent the unmixing problem as the following $l_2 - l_0$ optimization problem:

$$\min_X \frac{1}{2}\|Y - AX\|_F^2 + \lambda\|X\|_0 \quad \text{s.t. } X \geq 0 \tag{3}$$

where $\|\cdot\|_F$ is the Frobenius norm of the matrix and λ is the regularization parameter. Equation (3) is an NP-hard problem and very difficult to settle [38], [39]. In conventional sparse unmixing methods, the optimization problems can be expressed as follows:

$$\min_X \frac{1}{2}\|AX - Y\|_F^2 + \lambda\|X\|_{\rho,1} \quad \text{s.t. } X \geq 0. \tag{4}$$

1) By setting $\rho = 1$, we refer to (4) as SUnSAL [13], where $\|X\|_{1,1} = \sum_{j=1}^{n} \|x_j\|_1$ with x_j being the jth column of X. Based on the ADMM [36], SUnSAL can solve the corresponding optimization problem efficiently.
2) Similarly, $\rho = 2$ results in CLSUnSAL [17]. The term $\|X\|_{2,1} = \sum_{i=1}^{m} \|x^i\|_2$ is the $l_{2,1}$ mixed norm, which imposes sparsity to the lines of X. Here, x^i is the ith row of X. Different from SUnSAL, CLSUnSAL promotes joint sparsity among all the pixels.

III. SSMCSU Model

In this section, we describe the proposed SSMCSU model for sparse unmixing after spectral partitioning. We first introduce the two spectral partitioning techniques considered, i.e., AAP [31] and SD [34] spectral partitioning, and then provide details about the proposed MCSU model.

A. Multiview Data Construction

This step is to construct the multiview data set from the original HSI via spectral band partitioning. The original HSI is partitioned through two different considered techniques, i.e., AAP [31] and SD [34], which are driven by statistical principles and physical principles related to the imaging spectrometer, respectively. They can be applied to different hyperspectral data types. These partitions provide multiple views of the original HSI, and the generated multiview data set is used for MCSU.

Different from traditional clustering algorithms such as K-means [40] or ISODATA [41], Frey *et al.* [42] first propose the affinity propagation (AP) approach to automatically search for the optimal cluster number in the data. The AAP is presented to remove inconsistencies in the search process [31], and we can execute band clustering without knowing the number of exemplars in advance by using AAP. Let s be defined as the number of spectral partitions and c denote the number of band clusters. Following the work [32] for hyperspectral band clustering, the original HSI is divided into c-band clusters, each containing a set of highly similar spectral bands.

Considering that the collected spectra are usually a combination of more than one imaging spectrometer and different parts of the spectral signal have different characteristics, reference [34] develops an SD-based band partitioning strategy. The optical characteristics of a set of bands corresponding to

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each spectrometer share high correlation, and we can regard this as an alternative source for partitioning the spectral bands. Let q denote the number of spectrometers in a hyperspectral imaging system. The spectral groups are denoted by the spectral wavelengths available in each spectrometer, which is known in advance. Here, the q in SD is the same as c in AAP.

For the two band partitioning strategy, SD can be used for typical imaging spectrometers such as AVIRIS [43] and DAIS 7915 [44], and AAP can be used as a complement to other data types. After obtaining the q or c spectral groups (band clusters) through SD or AAP, we then reassign the bands with a high correlation into s spectral partitions by using an equal interval sampling band reassignment approach described in [34]. It should be noted that each partition contains spectral bands from different spectral groups (band clusters), which have lower similarity/correlation. The spectral similarity/correlation in the partitions is determined by the parameter s. In this way, spectral partitioning provides multiple views of the original image, and the spectral signature of each view follows the basic shape of the HSI, i.e., the spectral shapes of all views maintain consistency with that of the original HSI. This can be used to rectify the potential errors in the unmixing process and ensures that the data subset from different views contains the same set of endmembers. After spectral partitioning, we get s views of the original HSI $\{Y^{\upsilon}\}_{\upsilon=1}^{s}$, and $Y^{\upsilon} \in R^{L^{\upsilon}\times n}$ denotes the υth view of the HSI with L^{υ} spectral bands and n pixels, i.e., each view of the data subset contains the same original pixels but with different spectral bands.

B. SSMCSU

In order to explore the spectral information more effectively for sparse unmixing and inspired by the good potential of weighted sparse regularizer, an SMCSU method is proposed. In this way, the proposed SMCSU exploits the spectral correlation among different views of HSI via a weighting factor with less regularization parameters and good computational efficiency. Let $W_{\mathrm{ms}} \in R^{m\times n}$ be the spectral weighting matrix. The objective function of SMCSU is formulated as follows:

$$\min_{X^{\upsilon}} \sum_{\upsilon=1}^{s} \left(\frac{1}{2} \| A^{\upsilon} X^{\upsilon} - Y^{\upsilon} \|_F^2 + \lambda \| W_{\mathrm{ms}}^{\upsilon} \odot X^{\upsilon} \|_{\rho,1} \right) \quad \text{s.t. } X^{\upsilon} \geq 0 \tag{5}$$

where the operator $\odot$ denotes the elementwise multiplication of two variables, and λ is the regularization parameter. $A^{\upsilon} \in R^{L^{\upsilon}\times m}$ and $X^{\upsilon} \in R^{m\times n}$ correspond to the spectral library A and estimated abundance matrix X from the υth view, respectively. When $\rho \neq 1$ and $\rho \neq 2$, problem (5) is hard to solve [45]. In this article, we only focuses on the case of $\rho = 1$ or $\rho = 2$. Note that problem (4) is the special case of problem (5) when $s = 1$ [$W_{\mathrm{ms}} = I$ (the identity matrix)].

Candes *et al.* [18] prove that the weighted sparse regularizer can obtain a sparser result under a proper weighting factor. Clearly, how to design an effective weighting matrix to improve the sparsity of (5) is a crucial problem [18], [26]. Due to the influence of noise as well as the high mutual coherence in library A, the estimated abundances are often inaccurate and variable, thus making sparse unmixing unfaithful [15], [46]. For a multiview data model, the data subsets from different views obtained by spectral band partitioning have different characteristics, such as signal-to-noise ratio (SNR) [34]. This leads to a different unmixing performance on different views of the data set, which is reflected in the estimated abundance matrix.

To shed light on this issue, we use a simulated data set provided in [15] with nine endmember signatures randomly selected from a spectral library, which contains 50 mineral spectra from the USGS library [14]. The data set is corrupted by independent identically distributed Gaussian noise, and the SNR is set to 25 dB which makes the problem very difficult. The number of views is set to three after spectral partitioning using the SD spectral partitioning strategy.

Fig. 1 shows the unmixing results on different views of the data set with two selected endmembers. We can see that the estimated abundance matrix is unfaithful under the high level of noise. The endmember 17 is a true endmember that exists in the data and 31 is a virtual endmember produced due to the noise. For true endmember 17, the unmixing results (the abundance maps) from views 1–3 are consistent and similar to the true abundance map. However, the results of virtual endmember 31 are quite different on different views. That is to say, the abundance estimation of virtual endmembers from different views is easily dominated by noise, while true endmembers are not susceptible to noise. After constructing the multiview data set, the data subsets under different views contain different spectral bands, and different spectral bands have different levels of sensitivity to noise. As a result, the abundance estimation of the same virtual endmember has a different performance in different views. As we can see, the unmixing results of endmember 31 in views 1 and 2 are satisfactory, but the unmixing result in view 3 is poor. All of the endmembers both true and virtual show similar results as described above.

Based on these observations, we can distinguish between true and virtual endmembers by calculating the similarity of the abundance vectors belonging to the same endmember from different views. The less similar the abundance vectors of endmembers from different views, the less likely they are to be true endmembers. This is due to the fact that the data subsets of different views have a different noise sensitivity and endmembers with less similarity on the abundance vectors should be removed during the unmixing process. Utilizing this information and considering the rule of thumb that the value of the weight should be inversely proportional to the value of X^{υ} [18], we propose the following weighting matrix:

$$W_{\mathrm{ms}}^{\upsilon(t+1)}(i,:) = \frac{1}{(s-1)} \sum_{u=1,u\neq\upsilon}^{s} \frac{1}{f\left(X^{u,\upsilon(t)}(i,:)\right)+\sigma}. \tag{6}$$

where $\sigma > 0$ is a stabilization parameter and $W_{\mathrm{ms}}^{\upsilon(t+1)}(i,:)$ represents the ith row of $W_{\mathrm{ms}}^{\upsilon}$ in the $(t+1)$th iteration, where $f(\cdot)$ is a function explicitly exploiting the spectral correlations of different views by endmember similarity calculation. Note that the function $f(\cdot)$ can be any similarity measurement. For example, the Euclidean distance $E(i) = ((x_i^u - x_i^{\upsilon})(x_i^u - x_i^{\upsilon})^T)^{1/2}$, the Dice coefficient

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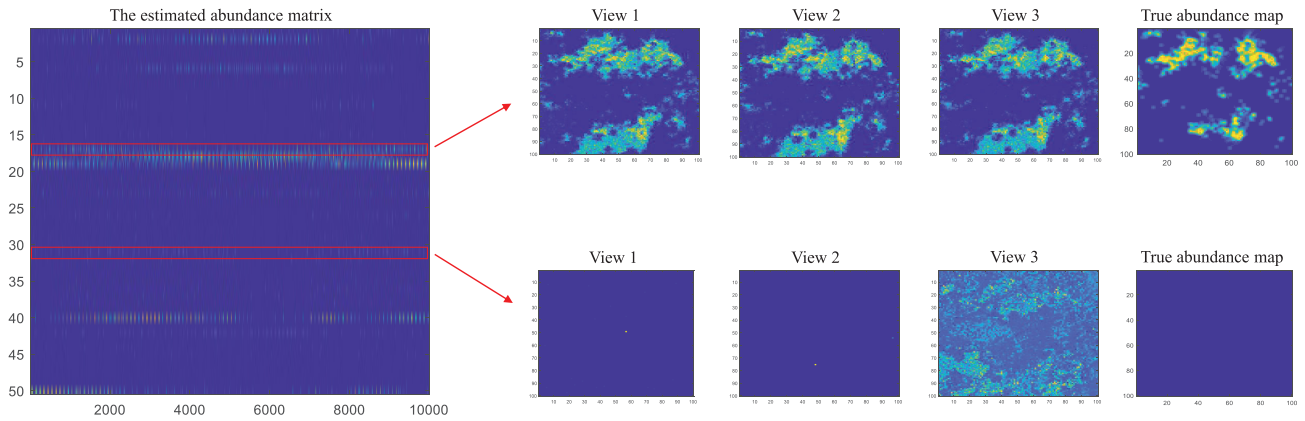


Fig. 1. Experimental illustration of the unmixing performance on multiview data set with two selected endmembers. The endmember 17 is a true endmember that exists in the data and 31 is a virtual endmember produced due to the noise.

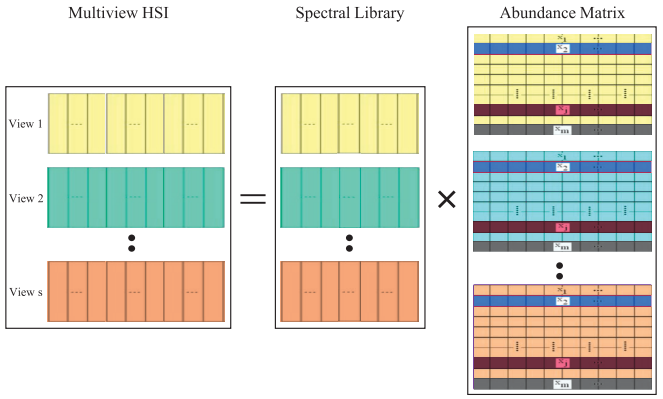


Fig. 2. Graphical illustration of the performance of weighted multiview collaborative regularizer with $\rho = 1$. It promotes row sparsity collaboratively in all views and penalizes the nonzero coefficients in the columns.

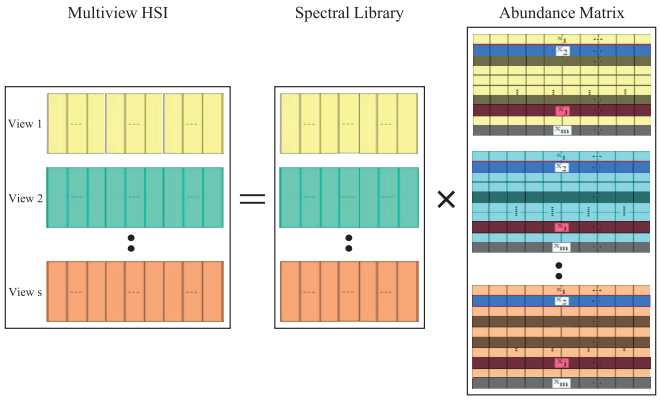


Fig. 3. Graphical illustration of the performance of double-collaborative sparse regularizer. Dark color lines indicate the possible generated virtual endmembers in different views. The regularizer can reduce these virtual endmembers through another collaborative in all views to find the true endmembers, which are represented as x_2, x_j, and x_m.

$D(i) = (2|x_i^u \cdot x_i^v|)/(|x_i^u|_2^2 + |x_i^v|_2^2)$, and the cosine formula $C(i) = (x_i^u x_i^v)/(|x_i^u||x_i^v|)$. Specifically, in this article, we use the spectral angle distance as the similarity measurement as follows:

$$f\left(X^{u,v}(i,:)\right) = \arccos\left(\frac{X^u(i,:)X^v(i,:)^T}{\|X^u(i,:)\|_2\|X^v(i,:)\|_2}\right). \quad (7)$$

In (6), the multiview spectral weighting factor $W_{\text{ms}}^v \in R^{m\times n}$ is presented in the form of a row matrix, with each row vector being inversely proportional to the similarity measure value of the corresponding abundance vector from different views estimated from the previous iteration. It is suggested that large weights are used to discourage nonzero row vectors in the recovered abundance matrix from different views simultaneously, whereas small weights encourage nonzero row vectors. That is, it applies joint sparsity among all the pixels in a multiview collaborativity manner.

Figs. 2 and 3 illustrate the effect of the multiview spectral-weighted sparse regularizer $\lambda\|W_{\text{ms}}^v \odot X^v\|_{\rho,1}$ imposing sparsity among the endmembers collaboratively for all views. By setting $\rho = 1$, as presented in Fig. 2, $\lambda\|W_{\text{ms}}^v \odot X^v\|_{1,1}$ introduces a prior with spectral sparsity under the l_1 framework with W_{ms}^v promoting nonzero row vectors in different views simultaneously. As such, the weighted l_1 norm will promote the sparsity of nonzero abundance vectors in all views collaboratively as well as penalize the nonzero coefficients in the columns. On the other hand, $\rho = 2$ as presented in Fig. 3, $\lambda\|W_{\text{ms}}^v \odot X^v\|_{2,1}$ provides a double-collaborative sparse regularizer with the $l_{2,1}$ mixed norm. It will promote both the sparsity of nonzero abundance vectors collaboratively in all views and the sparsity among the endmembers simultaneously for all pixels in a single view. Tests have shown that the solution of problem (4) with $\rho = 2$ cannot keep sparse all the time; it could generate more virtual endmembers, which are not active in the HSI [47], [48]. The proposed double-collaborative sparse regularizer can reduce virtual endmembers through another collaborative in all views to get a more accurate solution.

From (5), it is clear that the SMCSU model is a spectral-based HU method. Considering the importance of spatial information for sparse unmixing [15], [23], we integrate another weighting factor into the MCSU model to promote the piecewise smooth structure of the abundances. Therefore, we have the following SSMCSU model:

$$\min_{X^v} \sum_{v=1}^{s} \left(\begin{array}{l} \frac{1}{2}\|A^v X^v - Y^v\|_F^2 \\ +\lambda\|W_{\text{ms}}^v \odot W_{\text{spa}}^v \odot X^v\|_{\rho,1} \end{array} \right)$$
$$\text{s.t. } X^{(v)} \geq 0. \quad (8)$$

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Relying on the success of [23], $W_{\text{spa}}^{\upsilon}$ is updated at iteration $t+1$ as

$$W_{\text{spa}}^{\upsilon(t+1)}(i,j)=\frac{1}{f_{h\in\mathcal{N}(j)}\left(x_{ih}^{\upsilon(t)}\right)+\delta} \tag{9}$$

where $f(x_{ij}^{\upsilon})=(\sum_{h\in\mathcal{N}(j)}\varepsilon_{ih}x_{ih}^{\upsilon})/(\sum_{h\in\mathcal{N}(j)}\varepsilon_{ih})$, which exploits the spatial correlations through the neighborhood system $\mathcal{N}(j)$. The difference from [23] is that the spatial weight $W_{\text{spa}}^{\upsilon}$ is calculated separately in different views. The neighboring coverage considered here is the 8-connected (3×3 window) and $\varepsilon_{ij}=(1/((a-c)^2+(b-d)^2)^{1/2})$ represents the neighborhood importance for $x_i(a,b)$ and $x_j(c,d)$.

Finally, we will get s abundance estimation matrices corresponding to the s-view data set. How to calculate the final abundance matrix is a challenging and interesting problem, which is related to the multiview fusion techniques. Ammanouil *et al.* [33] studied and provided four different fusion methods and all of them are effective. For simplicity and fairness, in this article, we use the sample averaging method to calculate the final abundance matrix X as follows:

$$X=\sum_{\upsilon=1}^{s}X^{\upsilon}\Big/s. \tag{10}$$

To solve (5) and (8), we adopt the ADMM [36]. Following [18], [22] and [23], the corresponding problems can be iteratively solved efficiently by an outer–inner looping scheme. Given the weights W_{ms}^{υ} and $W_{\text{spa}}^{\upsilon}$, the coefficients X^{υ} can be iteratively and collaboratively optimized via the ADMM. Taking SSMCSU in problem (8) with $\rho=1$ as an example, here, we revisit the algorithm in detail to apply it to multiview scenes. Specifically, the optimization problem (8) can be equivalently expressed as follows for view υ:

$$\begin{aligned}\min_{U^{\upsilon},V_1^{\upsilon},V_2^{\upsilon},V_3^{\upsilon}}&\left(\begin{array}{l}\frac{1}{2}\|V_1^{\upsilon}-Y^{\upsilon}\|_F^2\\ +\lambda\|W_{\text{ms}}^{\upsilon}\odot W_{\text{spa}}^{\upsilon}\odot V_2^{\upsilon}\|_{1,1}+\iota_{R+}\left(V_3^{\upsilon}\right)\end{array}\right)\\ &\text{s.t. } V_1^{\upsilon}=A^{\upsilon}U^{\upsilon},\ V_2^{\upsilon}=U^{\upsilon},\ V_3^{\upsilon}=U^{\upsilon}\end{aligned} \tag{11}$$

where $\iota_{R+}(X^{\upsilon})=\sum_{j=1}^{n}\iota_{R+}(x_j^{\upsilon})$ is the indicator function, and $\iota_{R+}(x_j^{\upsilon})=0$ if x_j^{υ} belongs to the nonnegative orthant; otherwise, $\iota_{R+}(x_j^{\upsilon})=+\infty$. Then, optimization (11) can be further written in a compact form

$$\min_{U^{\upsilon},V^{\upsilon}} g(V^{\upsilon})\quad \text{s.t. } GU^{\upsilon}+BV^{\upsilon}=0 \tag{12}$$

where $V^{\upsilon}=(V_1^{\upsilon},V_2^{\upsilon},V_3^{\upsilon})$, $g(V^{\upsilon})=(1/2)\|V_1^{\upsilon}-Y^{\upsilon}\|_F^2+\lambda\|W_{\text{ms}}^{\upsilon}\odot W_{\text{spa}}^{\upsilon}\odot V_2^{\upsilon}\|_{1,1}+\iota_{R+}(V_3^{\upsilon})$, and $G=[A^{\upsilon},I,I]^T$, $B=\text{diag}(-I)$. Introducing the scaled Lagrange multipliers $D^{\upsilon}=(D_1^{\upsilon},D_2^{\upsilon},D_3^{\upsilon})$, the augmented Lagrange function associated with optimization (12) is written as follows:

$$\begin{aligned}&\mathcal{L}(U^{\upsilon},V^{\upsilon},D^{\upsilon})\\ &\equiv g(U^{\upsilon},V^{\upsilon})+\frac{\mu}{2}\|GU^{\upsilon}+BV^{\upsilon}-D^{\upsilon}\|_F^2\\ &=\frac{1}{2}\|V_1^{\upsilon}-Y^{\upsilon}\|_F^2+\lambda\|W_{\text{ms}}^{\upsilon}\odot W_{\text{spa}}^{\upsilon}\odot V_2^{\upsilon}\|_{1,1}+\iota_{R+}\left(V_3^{\upsilon}\right)\\ &\quad+\frac{\mu}{2}\|A^{\upsilon}U^{\upsilon}-V_1^{\upsilon}-D_1^{\upsilon}\|_F^2+\frac{\mu}{2}\|U^{\upsilon}-V_2^{\upsilon}-D_2^{\upsilon}\|_F^2\\ &\quad+\frac{\mu}{2}\|U^{\upsilon}-V_3^{\upsilon}-D_3^{\upsilon}\|_F^2\end{aligned} \tag{13}$$

Algorithm 1 Proposed SSMCSU Algorithm

Initialization:
set $k,t=0$, choose $\mu,\lambda,\sigma,\delta,\theta>0$, $U^{\upsilon(0)}$, $V_1^{\upsilon(0)}$, $V_2^{\upsilon(0)}$, $V_3^{\upsilon(0)}$, $D_1^{\upsilon(0)}$, $D_2^{\upsilon(0)}$, $D_3^{\upsilon(0)}$, $\upsilon=1,\ldots,s$.
For all the s views:
repeat
$W_{ms}^{\upsilon(t)}(i,:)\leftarrow\frac{1}{(s-1)}\sum_{u=1,u\neq\upsilon}^{s}\frac{1}{f((U^{\upsilon(t)}-D_2^{\upsilon(t)})^{u,\upsilon}(i,:))+\sigma}$, $i=1,\ldots,m$
$W_{spa}^{\upsilon(t)}(i,j)\leftarrow\frac{1}{f_{h\in\mathbb{N}(j)}((u_{ih}^{\upsilon(t)}-d_{2ih}^{\upsilon(t)})^{\upsilon})+\delta}$, $i=1,\ldots,m$ $j=1,\ldots,n$
repeat
$U^{\upsilon(k+1)}\leftarrow(A^{\upsilon T}A^{\upsilon}+2I)^{-1}(A^{\upsilon T}\xi_1+\xi_2+\xi_3)$, $\xi_1=V_1{}^{\upsilon(k)}+D_1{}^{\upsilon(k)}$, $\xi_2=V_2{}^{\upsilon(k)}+D_2{}^{\upsilon(k)}$, $\xi_3=V_3{}^{\upsilon(k)}+D_3{}^{\upsilon(k)}$
$V_1{}^{\upsilon(k+1)}\leftarrow\frac{1}{1+\mu}[Y^{\upsilon}+\mu(A^{\upsilon}U^{\upsilon(k+1)}-D_1{}^{\upsilon(k)})]$
$V_2{}^{\upsilon(k+1)}\leftarrow\text{soft}(U^{\upsilon(k+1)}-D_2{}^{\upsilon(k)},\frac{\lambda}{\mu}W_{ms}^{\upsilon(t)}W_{spa}^{\upsilon(t)})$
$V_3{}^{\upsilon(k+1)}\leftarrow\max(U^{\upsilon(k+1)}-D_3{}^{\upsilon(k)},0)$
Update Lagrange multipliers:
$D_1{}^{\upsilon(k+1)}\leftarrow D_1{}^{\upsilon(k)}-A^{\upsilon}U^{\upsilon(k+1)}+V_1{}^{\upsilon(k+1)}$
$D_2{}^{\upsilon(k+1)}\leftarrow D_2{}^{\upsilon(k)}-U^{\upsilon(k+1)}+V_2{}^{\upsilon(k+1)}$
$D_3{}^{\upsilon(k+1)}\leftarrow D_3{}^{\upsilon(k)}-U^{\upsilon(k+1)}+V_3{}^{\upsilon(k+1)}$
Update iteration: $k\leftarrow k+1$
$U^{\upsilon(t+1)}\leftarrow U^{\upsilon(k+1)}$
$D_2^{\upsilon(t+1)}\leftarrow D_2^{\upsilon(k+1)}$
Update iteration: $t\leftarrow t+1$
until $\|GU^{\upsilon(k)}+BV^{\upsilon(k)}\|_F<\theta$ or the maximum iteration is reached.

where $\mu>0$ is a positive constant. The ADMM iteratively minimizes $\mathcal{L}(U^{\upsilon},V^{\upsilon},D^{\upsilon})$ with respect to U^{υ} and V^{υ} followed by updating D^{υ} in the three consecutive steps:

$$\begin{cases}U^{\upsilon(k+1)}\leftarrow\arg\min\limits_{U^{\upsilon}}\mathcal{L}\left(U^{\upsilon},V^{\upsilon(k)},D^{\upsilon(k)}\right)\\ V^{\upsilon(k+1)}\leftarrow\arg\min\limits_{V^{\upsilon}}\mathcal{L}\left(U^{\upsilon(k+1)},V^{\upsilon},D^{\upsilon(k)}\right)\\ D^{\upsilon(k+1)}\leftarrow D^{\upsilon(k)}-GU^{\upsilon(k+1)}-BV^{\upsilon(k+1)}\end{cases} \tag{14}$$

until some stopping criterion is satisfied. The optimization problem involved in the SSMCSU algorithm is shown in Algorithm 1, where $\text{soft}(U,\tau)$ is the componentwise application of the soft-threshold function $u\rightarrow\text{sign}(u)\max\{|u|-\tau,0\}$.

IV. Experiments and Analysis

In this section, we explore the effectiveness of the proposed MCSU model for sparse unmixing on both simulated and real hyperspectral data. The results obtained by SUnSAL [13], SUnSAL-TV [15], DRSU [20], RMSU [35], and S^{2}WSU-W1 [23] algorithms are reported for comparative purposes. Several performance discriminators are used to substantiate the accuracy of unmixing process. For the reconstruction quality of spectral mixtures, the signal-to-reconstruction error (SRE) is adopted in this article to evaluate the unmixing accuracy. Let x be the true abundance and $\widehat{x}$ be the estimated abundance. The SRE is given as follows:

$$\text{SRE}\equiv E\left(\|x\|_2^2\right)/E\left(\|x-\widehat{x}\|_2^2\right) \tag{15}$$

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which is measured in decibels: SRE $\equiv 10\log_{10}(\text{SRE})$, where $E(\cdot)$ denotes the expectation function. Its larger values represent a higher quality unmixing result. We also compute the "probability of success," i.e., p_s, which indicates the probability that the relative error power is smaller than a threshold. It is defined as $p_s \equiv P(\|x - \widehat{x}\|_2/\|x\|_2 \leq \text{threshold})$. In this article, the threshold is 3.16 (5 dB), which is demonstrated to be appropriate in [13]. Furthermore, the sparsity, which measures the proportion of the number of elements in $\widehat{x}$ that is greater than 0.005 in all elements, is also introduced. Obviously, its smaller values indicate a more sparse result.

A. Data Sets

Two spectral libraries of minerals extracted from the USGS library [14] are considered in our experiments. The spectral library A_1 contains 240 materials, and the library A_2 contains a selection of 498 materials. The reflectance values are measured for 224 spectral bands distributed uniformly in the range 0.4–2.5 μm. Two different data sets are simulated, and their fractional abundances are subject to the ANC and ASC constraints.

1) Simulated Data Cube 1 (DC1): DC1 is generated with 100×100 pixels and provided by Iordache *et al.* [15]. The fractional abundances are smooth with sharp transitions as illustrated in Fig. 4, which shows the true fractional abundance of nine endmember signatures. They are randomly chosen from the spectral library A_1. The data set is corrupted by independent identically distributed Gaussian noise with different levels (SNR $=$ 20, 30, and 40 dB).

2) Simulated Data Cube 2 (DC2): DC2 is an image of size 64×64 pixels, which is provided by Tang *et al.* [49]. This data set also exhibits a spatial homogeneity, and the true fractional abundance of five endmembers is shown in Fig. 5. The spectral library A_2 is used in this data. There is no pure pixel included in the data set. Similar to DC1, different levels of Gaussian noise (SNR $=$ 20, 30, and 40 dB) is included in order to test the robustness of algorithms.

3) Real Hyperspectral Data 1: The first real data used in the experiment are the Jasper Ridge data set, which is captured by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor. There are 224 spectral bands recorded in each pixel ranging from 0.38 to 2.5 μm. Prior to the analysis, low SNR and water-vapor absorption bands (bands 1–3, 108–112, 154–166, and 220–224) are removed before unmixing. As used in [12], [50], and [51], the subregion of the original data set containing 100×100 pixels with 198 spectral bands is considered here, which is shown in Fig. 6(a). Four endmembers: Tree, Water, Soil, and Road are observed in the Jasper Ridge data set.

4) Real Hyperspectral Data 2: The real data 2 used is the well-known AVIRIS Cuprite data set, which is a common benchmark to evaluate the performance of unmixing algorithms [15], [49], [52]. The Cuprite data comprise 224 bands, covering the wavelength range of 0.4–2.5 μm. Similarly, blank bands (bands 1 and 2, 105–115, 150–170, and 223 and 224) are removed before unmixing, remaining 188 spectral bands. The portion which corresponds to a 250×191 pixels subset of the Cuprite data set as shown in Fig. 7 is used in the experiment.

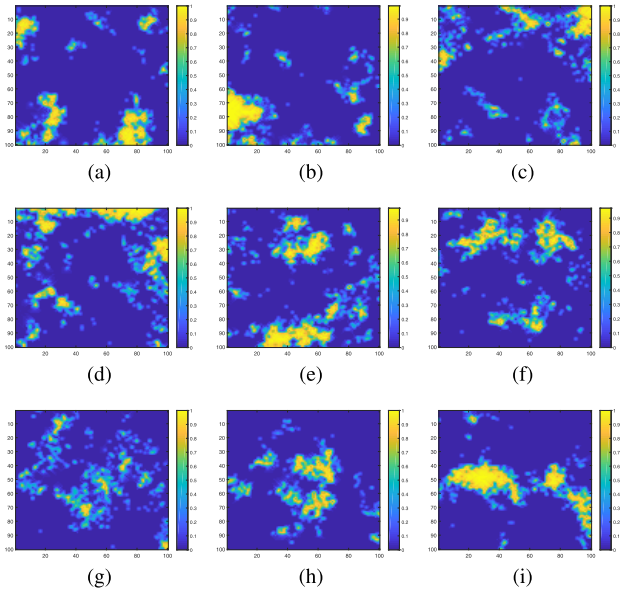


Fig. 4. True fractional abundances of DC1. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4. (e) Endmember 5. (f) Endmember 6. (g) Endmember 7. (h) Endmember 8. (i) Endmember 9.

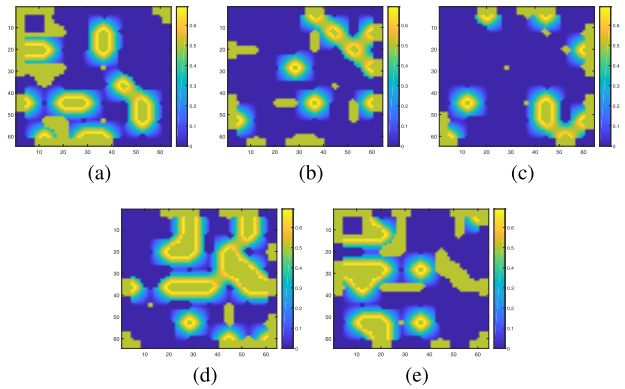


Fig. 5. True fractional abundances of DC2. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4. (e) Endmember 5.

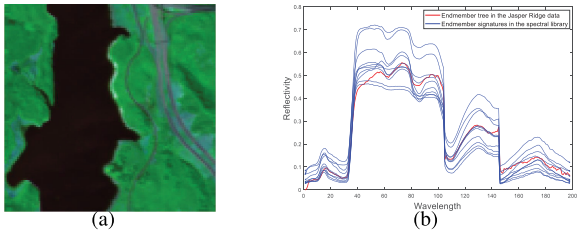


Fig. 6. (a) Jasper Ridge data set with bands 10, 90, and 180. (b) Endmember Tree selected in the Jasper Ridge data set (red) and some endmembers in the USGS spectral library A_2 (blue).

B. Experiments on the MCSU Model

1) Test of the Convergence: The weighted problem is often optimized via the ADMM with an outer–inner looping scheme,

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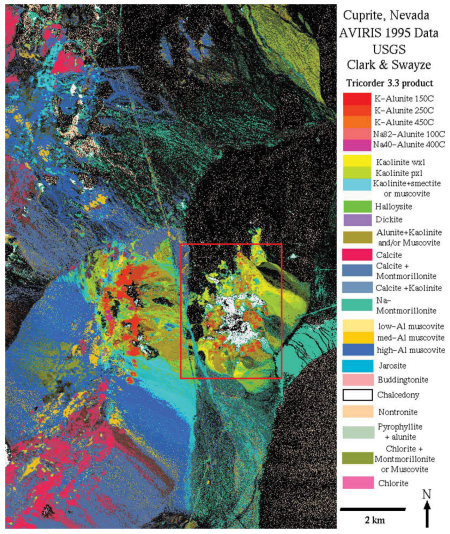


Fig. 7. Distribution of different minerals for Cuprite data obtained by Tricorder software. The subimage enclosed by the red rectangle is used.

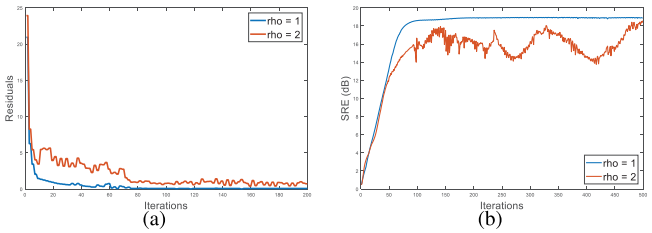


Fig. 8. Test of the convergence for the complete algorithm. (a) Residual $\|\mathrm{GU}^{v(k)} + \mathrm{BV}^{v(k)}\|_F$ as a function of the number of iterations. (b) SRE as a function of the number of iterations.

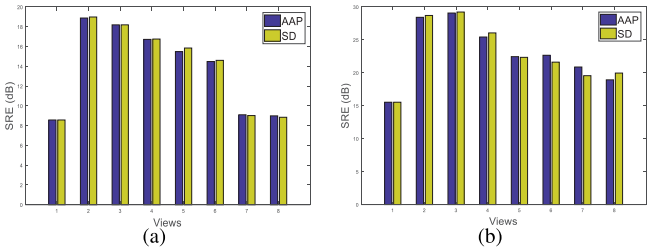


Fig. 9. Test of different spectral partitioning strategies. (a) Results of different partitioning strategies on DC1 (SNR = 30 dB). (b) Results of different partitioning strategies on DC2 (SNR = 30 dB).

which has obtained with very promising performance [18], [22], [23]. The algorithm stops when $\|\mathrm{GU}^{v(k)} + \mathrm{BV}^{v(k)}\|_F < \theta$ or the maximum iteration is reached in practice, which is quite common. Fig. 8(a) and (b) shows the obtained residual, i.e., $\|\mathrm{GU}^{v(k)} + \mathrm{BV}^{v(k)}\|_F$, and SRE (dB) as a function of iterations for the proposed SMCSU algorithm, respectively, using DC1 with 30-dB SNR. The number of iterations is set to 5 and 200, respectively, for the inner and outer loops.

It can be seen that the complete scheme results in good convergence when $\rho = 1$. However, when $\rho = 2$, the full scheme shows a less satisfactory result, with an oscillation in the convergence process despite being at a relatively high level of SRE value. This indicates that the weighted l_1

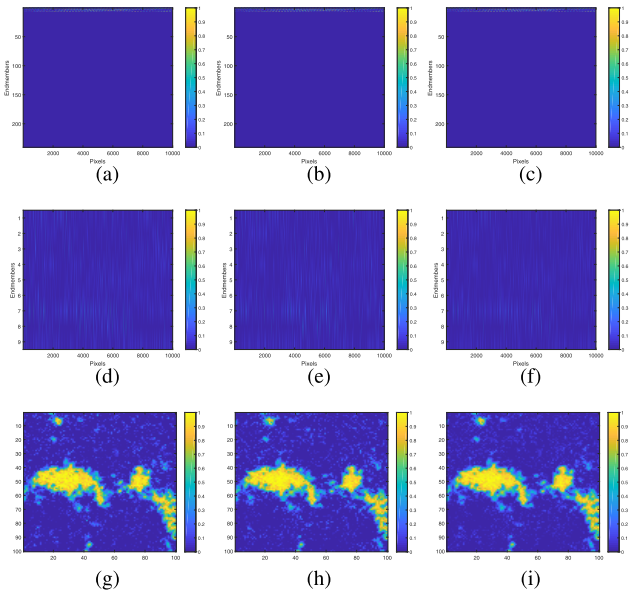


Fig. 10. Estimated abundances obtained by SSMCSU for each endmember from (a) view 1, (b) view 2, and (c) fusion (AVG result), on DC1 (SNR = 20 dB). Difference abundances for the nine true endmembers between the true abundances and the estimated result obtained from (d) view 1, (e) view 2, and (f) fusion (AVG result). Abundance maps obtained for the ninth endmember from (g) view 1, (h) view 2, and (i) fusion (AVG result).

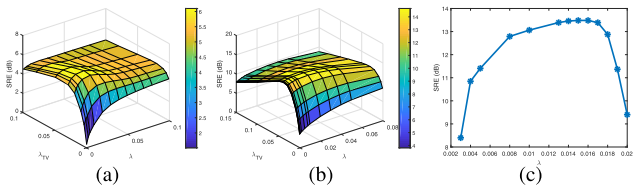


Fig. 11. SRE as a function of parameters λ or/and λ_{TV} (SNR = 20 dB). (a) Results of SUnSAL-TV on DC1. (b) Results of SUnSAL-TV on DC2. (c) Results of SSMCSU on DC1.

minimization is superior to the weighted l_2 minimization in the process of recovering sparse signals under the proposed W_{ms} weighting factor. Candes *et al.* [18] attributed this fact to the natural tendency of l_1 minimization to encourage sparsity. An in-depth study of the reweighting strategies for l_2 minimization is out of the scope of this article [53], [54], and this again indicates that designing an effective weighting factor is a crucial problem. Therefore, in our experiments performed in this article, the MCSU model used is under the weighted l_1 minimization, i.e., $\rho = 1$, with the number of iterations involved in k and t are set to 5 and 200, respectively.

2) Test of Different Spectral Partitioning Strategies: With the introducing of the spectral partitioning techniques, how to select the effective partitioning strategies and choose the optimal number of views becomes an issue. In order to illustrate this problem, Fig. 9(a) and (b) shows the experiments on different partitioning strategies with different number of views, i.e., s, from 1 to 8. We find that the results obtained by these two strategies show the same trend, and the SD partitioning strategy is slightly better than AAP in the vicinity of the best results. On the other hand, the best result

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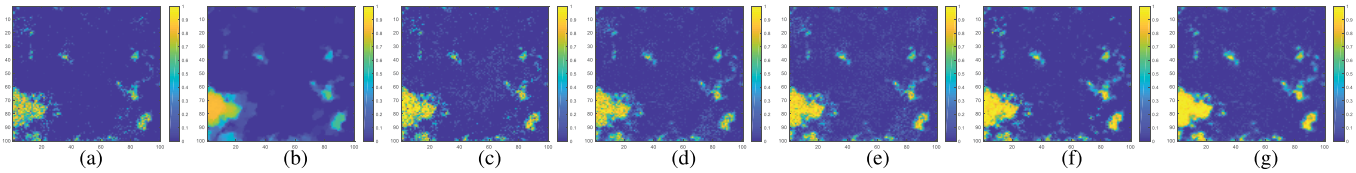


Fig. 12. Abundance maps for the second endmember from DC1 with SNR = 20 dB obtained by (a) SUnSAL, (b) SUnSAL-TV, (c) DRSU, (d) RMSU, (e) SMCSU, (f) S^2WSU-W1, and (g) SSMCSU.

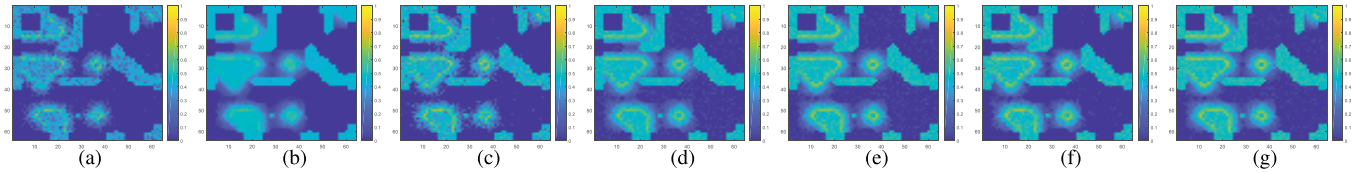


Fig. 13. Abundance maps for the fifth endmember from DC2 with SNR = 20 dB obtained by (a) SUnSAL, (b) SUnSAL-TV, (c) DRSU, (d) RMSU, (e) SMCSU, (f) S^2WSU-W1, and (g) SSMCSU.

is at $s = 2$ or $s = 3$. In practice, relatively good results can be obtained at both values.

We note that the spectral partitioning techniques can reduce spectral similarities while increasing the diversities among partitions. However, as the number of partitions increases, i.e., the number of views increases, the spectral bands contained in each subset of the multiview data set are reduced, which affects the regression process of the model. Therefore, there is a balance among the values of s, which is at $s = 2$ or $s = 3$. It should be finally noted that with the development of remote sensing technology, the spectral resolution of imaging spectrometers is getting higher and higher, and the spectral bands are more and more, such as ultraspectral imaging. Under such a trend, spectral partitioning, or multiview spectral analysis, will exert more and more potential. Based on the results shown in Fig. 9, in our experiments performed in this article, we adopt the SD strategy by setting $s = 2$ for DC1 and $s = 3$ for DC2, respectively. For the two real hyperspectral data sets, the value of s is also set to 2.

3) Test of Different Abundance Fusion Strategies: To calculate the final abundance matrix from the s abundance estimation matrices is challenging and interesting. During the optimization process, the abundance estimations from different views are iteratively and collaboratively optimized through the proposed weighting matrix. The resulting s abundance estimation matrices are similar. We note that any of the s matrices can be used as the final abundance estimation matrix. However, we can achieve a relatively better abundance estimation result through the multiview fusion technology. Here, the abundance fusion formula used is given as follows:

$$x = \sum_{v=1}^{s} \left(\frac{1}{d_v} x^v \right) \Big/ \sum_{v=1}^{s} \frac{1}{d_v} \tag{16}$$

where d_v is the metric value. Following [33], we study the following four different distance metrics.

1) *Simple Averaging (AVG):*

$$d_v = 1 \quad \forall v. \tag{17}$$

TABLE I

SRE SCORE ACHIEVED BY THE MCSU MODEL USING DIFFERENT ABUNDANCE FUSION STRATEGIES ON DC1 AND DC2 (SNR = 20 dB)

Data set	Algorithm	view 1	view 2	view 3	AVG	ANG	MSE	MAD
DC1	SMCSU	9.8517	9.8539	/	10.5649	10.5691	10.5731	10.5690
	SSMCSU	12.8866	12.8510	/	13.4816	13.4817	13.4896	13.4816
DC2	SMCSU	18.4158	18.2132	18.3939	19.3870	19.4050	19.4124	19.4046
	SSMCSU	20.0517	19.9058	19.9319	21.0716	21.0778	21.0820	21.0771

2) *The Reconstruction Angle (ANG):*

$$d_v = \cos^{-1}\left(\langle y, \widehat{y}^v \rangle / \|y\| \|\widehat{y}^v\| \right). \tag{18}$$

3) *The Mean Square Error (MSE):*

$$d_v = \frac{1}{L} \sum_{j=1}^{L} \left(\widehat{y}_j^v - y_j \right)^2. \tag{19}$$

4) *The Mean Absolute Difference (MAD):*

$$d_v = \frac{1}{L} \sum_{j=1}^{L} |\widehat{y}_j^v - y_j| \tag{20}$$

where $\widehat{y}^v$ is the reconstructed pixel corresponding to view v.

Table I shows the SRE (dB) results obtained by the proposed approaches on DC1 and DC2, with 20-dB SNR, using different abundance fusion strategies. We can see that the SRE values achieved on different views are similar. After using abundance fusion strategies, the SRE values have a relative improvement. Meanwhile, the results on four different distance metrics are not much different. For its simplicity, we use the simple averaging (AVG) to compute the final abundance matrix in this article, as (10). Fig. 10 visualizes the abundance matrices, difference abundances, and abundance maps (for endmember 9) achieved by SSMCSU on DC1 (SNR = 20 dB) from view 1, view 2, and fusion (AVG result), respectively. We can get the same conclusion as the aforementioned. The abundance matrices obtained from different views are similar, but there are differences in some pixels. We observe that the abundance matrices from different views are complementary,

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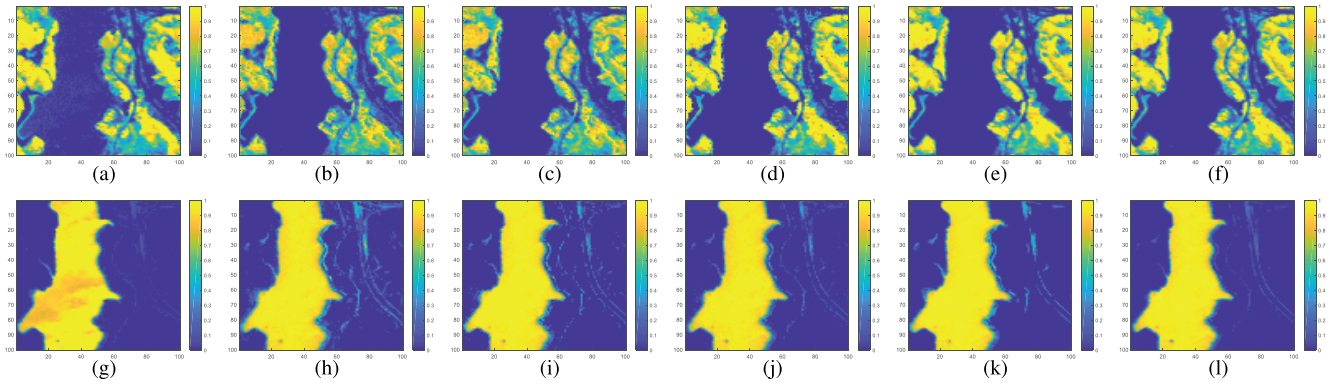


Fig. 14. Fractional abundance maps for the Jasper Ridge data set obtained by different unmixing methods. (Top row) Tree. (Bottom row) Water. (a) and (g) Ground truth. (b) and (h) DRSU. (c) and (i) RMSU. (d) and (j) SMCSU. (e) and (k) S^{2}WSU-W1. (f) and (l) SSMCSU.

TABLE II

SRE, p_s, AND SPARSITY SCORES ACHIEVED AFTER APPLYING DIFFERENT UNMIXING METHODS ON DC1

Algorithm	SNR = 20dB			SNR = 30dB			SNR = 40dB		
	SRE(dB)	p_s	*sparsity*	SRE(dB)	p_s	*sparsity*	SRE(dB)	p_s	*sparsity*
SUnSAL	4.2349 (λ=2e−1)	0.5546	0.0346	8.5714 (λ=2e−2)	0.7947	0.0496	15.1838 (λ=5e−3)	0.9907	0.0427
SUnSAL-TV	6.0098 (λ=2e−2; λ_{TV}=3e−2)	0.6466	0.0558	12.0168 (λ=1e−2; λ_{TV}=4e−3)	0.9519	0.0554	18.3969 (λ=5e−3; λ_{TV}=1e−3)	1	0.0340
DRSU	4.5250 (λ=1.5e−2)	0.6095	0.0335	14.7753 (λ=2e−3)	0.9699	0.0230	29.6890 (λ=1e−3)	1	0.0128
RMSU	10.0128 (λ=5e−1)	0.9123	0.0222	17.7928 (λ=3e−1)	1	0.0220	28.8035 (λ=3e−2)	1	0.0160
SMCSU	10.5649 (λ=1.5e−2)	0.9263	0.0243	18.9880 (λ=8e−3)	1	0.0211	31.2731 (λ=3e−3)	1	0.0123
S^{2}WSU-W1	7.9661 (λ=2e−2)	0.8161	0.0306	20.6822 (λ=5e−3)	1	0.0186	32.1356 (λ=3e−3)	1	0.0121
SSMCSU	**13.4816** (λ=1.5e−2)	**0.9693**	**0.0179**	**21.6671** (λ=5e−3)	1	**0.0162**	**33.0992** (λ=2e−3)	1	**0.0119**

TABLE III

SRE, p_s, AND SPARSITY SCORES ACHIEVED AFTER APPLYING DIFFERENT UNMIXING METHODS ON DC2

Algorithm	SNR = 20dB			SNR = 30dB			SNR = 40dB		
	SRE(dB)	p_s	*sparsity*	SRE(dB)	p_s	*sparsity*	SRE(dB)	p_s	*sparsity*
SUnSAL	7.2310 (λ=2e−1)	0.7959	0.0199	15.5317 (λ=3e−2)	1	0.0211	26.6143 (λ=8e−3)	1	0.0102
SUnSAL-TV	14.2797 (λ=2e−2; λ_{TV}=4e−2)	1	0.0143	22.4410 (λ=5e−3; λ_{TV}=1e−2)	1	0.0075	28.1812 (λ=1e−3; λ_{TV}=3e−3)	1	0.0125
DRSU	11.2857 (λ=2e−1)	0.9147	0.0127	28.1035 (λ=5e−2)	1	0.0057	37.5066 (λ=1e−2)	1	0.0050
RMSU	18.1431 (λ=8e−1)	1	0.0090	27.7009 (λ=5e−1)	1	0.0078	37.5070 (λ=3e−1)	1	0.0057
SMCSU	19.3870 (λ=2e−2)	1	0.0088	29.7473 (λ=2e−2)	1	0.0056	39.1219 (λ=1e−2)	1	0.0050
S^{2}WSU-W1	17.8492 (λ=1e−1)	1	0.0087	30.2405 (λ=2e−2)	1	0.0054	40.7768 (λ=1e−2)	1	0.0050
SSMCSU	**21.0716** (λ=2e−2)	1	**0.0070**	**30.9380** (λ=2e−2)	1	**0.0052**	**40.8911** (λ=5e−3)	1	0.0050

which makes the abundance fusion results have better accuracy. The better results obtained from fusion (AVG result) are also reflected on the abundance map of endmember 9.

C. Comparison of MCSU With Other Methods

1) Experiments on Simulated Data Cube: First, we discuss the affection of regularization parameters on the considered algorithms. Fig. 11(a)–(c) shows the achieved SRE (dB) as a function of regularization parameters on DC1 and DC2 (SNR = 20 dB) for the SUnSAL-TV and proposed SSMCSU algorithms. We can observe that the performance of SUnSAL-TV is very dependent on the setting of parameter λ_{TV}. For the proposed SSMCSU algorithm, it has only one parameter and is quite easy to achieve a suboptimal parameter setting. The effects of the regularization parameters on the other algorithms, i.e., SUnSAL, DRSU, SMCSU, and S^{2}WSU-W1, are similar to that of SSMCSU.

Tables II and III show the SRE (dB), p_s, and sparsity scores obtained by different tested approaches with DC1 and DC2, thus covering all considered SNR levels. The highest results achieved over the range of considered parameters are shown (the optimal regularization parameters are marked in parentheses). Note that SUnSAL, DRSU, and SMCSU are spectral-based methods, while SUnSAL-TV, S^{2}WSU-W1, and SSMCSU are spectral–spatial-based methods. RMSU is an efficient dictionary pruning algorithm. It can be seen from Tables II and III that the proposed SMCSU and SSMCSU algorithms, respectively, achieve better SRE (dB) and p_s scores than other spectral and spectral–spatial-based approaches in all the cases. Furthermore, the advantage of the multiview collaborative weighting factor is even higher. SRE (dB) and p_s achieved by SMCSU and SSMCSU are much better than those of SUnSAL, DRSU, RMSU, and SUnSAL-TV, and S^{2}WSU-W1 in the case of low SNR, which indicates that the inclusion of multiview spectral and spatial information results in high robustness. In addition, the sparsity performance improvements in SMCSU and SSMCSU indicate that the multiview spectral and spatial weighting factors can further enhance the sparsity. These findings clearly illustrate the potential of the proposed MCSU model.

For illustrative purposes, Figs. 12 and 13 exhibit a graphical comparison of the considered algorithms with only the abundance maps of endmember 2 for DC1 and endmember 5 for DC2 are displayed. The unmixing results achieved by SSMCSU are close to the ground truth abundances as shown in Figs. 4(b) and 5(e), which indicates great advantages over other approaches.

2) Experiments on Real Data 1: The reference ground truth of the Jasper Ridge data set is obtained in the same way as in [12] and [50]. The four endmember spectral signatures are manually selected from the image. Two ground truth abundance maps of the Jasper Ridge data set are shown in the first column of Fig. 14. The spectral library used for this hyperspectral data consists of 502 spectra, including four ground truth spectra in the scene and the mineral spectra in the spectral library A_2 (blank bands are removed). A large

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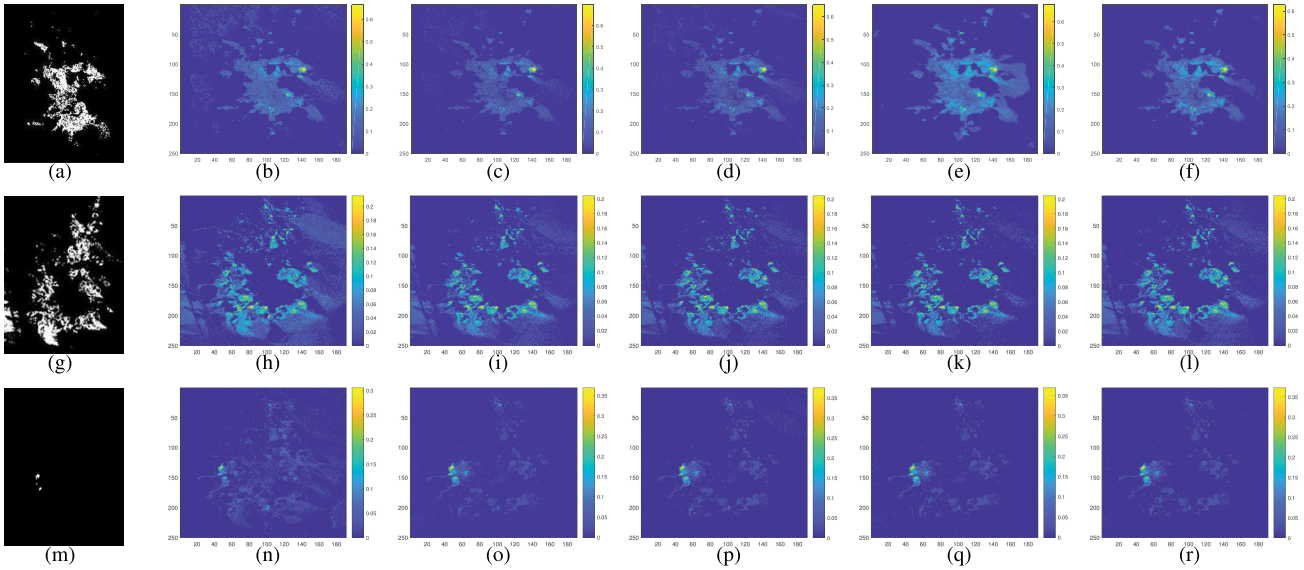


Fig. 15. Fractional abundance maps for the Cuprite data obtained by different unmixing methods and the distribution maps produced by Tetracorder 3.3 for the considered AVIRIS Cuprite scene. (From top to bottom) Chalcedony, Kaolinite, and Buddingtonite. (a), (g), and (m) Classification maps produced by Tricorder. (b), (h), and (n) DRSU. (c), (i), and (o) RMSU. (d), (j), and (p) SMCSU. (e), (k), and (q) S^{2}WSU-W1. (f), (l), and (r) SSMCSU.

TABLE IV

SRE, p_s, AND SPARSITY SCORES ACHIEVED AFTER APPLYING DIFFERENT UNMIXING METHODS ON THE JASPER RIDGE DATA SET

Algorithm	SUnSAL	SUnSAL-TV	DRSU	RMSU	SMCSU	S^{2}WSU-W1	SSMCSU
SRE(dB)	7.6115	8.2055	8.1851	8.8347	10.7127	11.2040	13.5567
p_s	0.8251	0.8549	0.8493	0.8682	0.9141	0.9308	0.9652
sparsity	0.0188	0.0217	0.0161	0.0123	0.0057	0.0061	0.0051

number of endmembers in the USGS are purposely added to the spectral library to investigate the ability of the sparse unmixing methods to distinguish the true endmembers. It is worth pointing out that there are many spectra similar to the four real endmembers in the constructed spectral library. For example, Fig. 6(b) shows the real endmember Tree and several endmember spectra similar to it in the spectral library. In this experiment, the parameters λ (λ_{TV}) used for SUnSAL, SUnSAL-TV, DRSU, RMSU, S^{2}WSU-W1, SMCSU, and SSMCSU are set to 0.002, 0.002 (0.001), 0.001, 0.02, 0.02, 0.01, and 0.02, respectively.

Table IV shows the SRE (dB), p_s, and sparsity scores achieved by different unmixing approaches with the reference ground truth abundances. It can be seen that the proposed MCSU approaches obtain better SRE (dB) and p_s scores than other spectral and spectral–spatial-based methods. This indicates the effectiveness of utilizing multiview spectral and spatial information in real HSI. Meanwhile, the lower sparsity values reveal that SMCSU and SSMCSU use fewer endmember spectral signatures to reconstruct the original image, that is, they can obtain a more sparse abundance matrix. Fig. 14 shows a qualitative comparison of the main tested algorithms with the abundance maps of endmember Tree and endmember Water are displayed. Compared with the reference ground truth, SSMCSU obtains the most similar abundance maps, which shows that the multiview spectral and spatial weights are effective and meaningful.

3) Experiments on Real Data 2: The spectral library used in this experiment is the A_1 library with the corresponding bands removed. Fig. 7 shows the mineral map produced by a Tricorder 3.3 software product [55], which is a good indicator for qualitative assessment of different unmixing algorithms. Because the ground true abundance maps of the Cuprite data are unavailable, we can only use the classification maps (the first column of Fig. 15) produced by Tricorder as a reference to make a qualitative analysis of the abundances achieved by different unmixing methods. Meanwhile, we adopt the RMSE instead of SRE to quantitatively evaluate the tested algorithms. The RMSE of an HSI is calculated by

$$\mathrm{RMSE} = \frac{1}{L}\sum_{j=1}^{L}\sqrt{\frac{\sum_{i=1}^{n}\left(y_{ji}-\widehat{y}_{ji}\right)^2}{n}}. \tag{21}$$

Fig. 15 shows a qualitative comparison among the abundance maps estimated by the main tested algorithms of three typical minerals: 1) Chalcedony; 2) Kaolinite; and 3) Buddingtonite. In this experiment, the parameters λ used for SUnSAL, DRSU, RMSU, S^{2}WSU-W1, SMCSU, and SSMCSU are empirically set to 0.001, 0.0001, 0.01, 0.002, 0.003, and 0.003, respectively, whereas the parameters for SUnSAL-TV are set to $\lambda = 0.001$ and $\lambda_{\mathrm{TV}} = 0.001$.

The classification map produced by Tricorder software is pixel level and clear-cut. However, the abundance map obtained by unmixing is subpixel level, and each pixel is a mixture of several minerals. Despite this, from Fig. 15, we can conclude that MCSU and other unmixing methods are all valid to cope with the Cuprite data set, with high abundances of the pixels indicating the presence of corresponding minerals. The abundance maps of tested sparse unmixing methods are similar; however, the dark regions with large abundance values distribute differently. It can be observed that the abundance

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TABLE V

RMSE ACHIEVED FOR THE CUPRITE DATA BY DIFFERENT UNMIXING METHODS

Algorithm	SUnSAL	SUnSAL-TV	DRSU	RMSU	SMCSU	S^2WSU-W1	SSMCSU
RMSE	0.0679	0.0585	0.0607	0.0597	0.0574	0.0602	0.0575

TABLE VI

COMPUTATIONAL TIMES IN SECONDS FOR THE CUPRITE DATA OBTAINED BY DIFFERENT UNMIXING METHODS

Algorithm	SUnSAL	SUnSAL-TV	DRSU	RMSU	SMCSU	S^2WSU-W1	SSMCSU
Time (s)	270.17	1533.25	303.53	120.62	439.39	447.43	861.94

maps achieved by DRSU and RMSU contain more noise than those of our methods, and the abundance maps yielded by DRSU are without good spatial consistency. The abundance maps estimated by SMCSU, S^2WSU-W1, and SSMCSU are generally consistent and have higher quality in comparison with DRSU and RMSU. S^2WSU-W1 and SSMCSU have better spatial consistency due to the addition of spatial weights.

Table V shows the RMSE values obtained by the tested algorithms. The RMSE of the proposed MCSU methods is the smallest, which demonstrates that the MCSU model has better unmixing performances. Meanwhile, the sparsity obtained by DRSU, RMSU, SMCSU, S^2WSU-W1, and SSMCSU are 0.0469, 0.0508, 0.0333, 0.0344, and 0.0322, respectively. This indicates that the proposed methods can achieve higher sparsity, i.e., the MCSU model uses a smaller number of elements to interpret the data set. Thus, the newly developed MCSU approaches offer the potential to improve the unmixing results in real analysis scenarios.

Finally, Table VI reports the processing time for the Cuprite data set. All the algorithms are implemented using MATLAB R2017a on a desktop PC equipped with an Intel Core 7 Quad CPU (at 4.00 GHz) and 16 GB of RAM. It can be seen that the pruning algorithm RMSU is the fastest, which utilizes a smaller spectral library. The proposed SMCSU and SSMCSU are slower than spectral-based (SUnSAL and DRSU) and spatial-based (S^2WSU-W1) methods, respectively, due to the increased model complexity, while the proposed MCSU model is faster than SUnSAL-TV benefiting from the selection of a weighted sparse regularizer.

V. CONCLUSION

In this article, we have developed a new MCSU method to make full use of the rich spectral and spatial information in HSI by the form of spectral- and spatial-weighted sparse regularizer. We put forward the idea of multiview sparse unmixing and utilize spectral partitioning techniques to construct the multiview data set. The corresponding optimization problem is iteratively solved by an outer–inner looping scheme with ADMM. The experimental results on both simulated and real hyperspectral data indicate that that the proposed MCSU model substantially improves the unmixing performance than other state-of-the-art unmixing methods. Despite the achieved very encouraging experimental results in this article, as with any new approach, there are some issues that may be challenging over time and worth future investigation. First, efficient parallel computing is needed to further optimize the new MCSU model for HU. Furthermore, the study of reweighting strategies for l_2 minimization in multiview sparse unmixing is still deserving of further investigation. Finally, further experiments on more data sets, especially higher resolution remote sensing images such as ultraspectral images, should be conducted to further evaluate the performance of MCSU.

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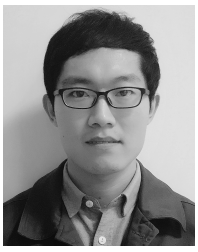
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